2026 MESA IDEA ACCELERATOR

Portable Personal Cooling Shelter

Bringing the cool environment to the worker.



Concept rendering based on the Fusion design

TEAM / CALCULATED VISIONS

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A Growing Workplace Crisis

Heat-related illness disrupts outdoor work

3,389

heat-related workplace injuries & illnesses

per year requiring days away from work

U.S. annual average, 2011–2020

Source: U.S. Bureau of Labor Statistics, SOII, 2011–2020 (average reported in team research)

Who We Designed For

Field crews and experiences behind the design

WHO HELPED SHAPE THE DESIGN

6+ worker interviews

1. Construction workers
2. Forest Service & conservation crews
3. Wildfire contractors

Interviewees reviewed ideas and gave feedback on design choices.

A FIELD EXPERIENCE THAT STUCK WITH US

6 of 12

crew members reportedly experienced heat illness by day six

A Forest Service worker recalled hiking into a remote worksite for an approximately 10-day job.

Interview account, not an independently verified case count

Source: Team worker interviews; Forest Service account is an interviewee recollection

The Design Challenge

What do existing heat-relief approaches leave unresolved?

EXISTING APPROACH	PRACTICAL LIMITATION
Shade canopies	Block the sun, but do not actively cool air
Cooling vests & wraps	Limited duration; some packs need re-cooling
Misting fans	Require water; less effective in humidity
Vehicle A/C & trailers	Cooling may be away from the worksite

THE CHALLENGE **Combine portability, enclosure, and active cooling near the worksite.**

Sources: Team interviews and market review; Cal/OSHA Title 8 §3395

Our Solution

Active. Portable. Cooling.



Portable Personal Cooling Shelter

Reflective enclosure

Insulation intended to reduce heat gain

Simple support frame

Simplified design for practical assembly

Modular active cooling

A/C concept for a small cool-down space

Concept rendering based on Fusion design; proposed features are not validated performance claims

Human-Centered Design in Action

Interviewee feedback helped turn field problems into design choices

WHAT WORKERS SHARED	HOW WE RESPONDED	WHAT COMES NEXT
Remote sites reached on foot	Compact, one-person shelter	Carry + packed-size checks
Heat illness over multiple days	Reflective shell + A/C concept	Thermal + power tests
Cooling breaks in vehicles	Cool-down space near work	Placement + user feedback

Workers gave feedback as we developed and refined the concept.

COMPLETED: Fusion CAD + 1:12 3D print | NEXT: full-size functional validation

Source: Team interviews and iterative concept feedback; functional testing is a next step

Market Potential & Next Steps

Start with the crews we interviewed; validate before expanding

INITIAL CUSTOMER FOCUS

Employers supporting remote outdoor crews

- Construction
- Forestry & conservation
- Wildfire contractors

Later opportunities Utilities, agriculture & other field crews

PROPOSED IMPLEMENTATION PATH

01

1:12 physical model

Check geometry, fit and component placement

02

Functional cooling prototype

Measure temperature change and power demand

03

Field feedback & refinement

Evaluate carryability and assembly on site

Potential market and testing phases are proposed; no field trials or commercial performance established

What We Learned & What Comes Next

01

Co-design with workers

Interviewee feedback shaped our design choices.

02

Keep the design practical

Simple construction matters in field conditions.

03

Move toward validation

Full-size testing is our next engineering milestone.

NEXT MILESTONE **Secure resources for full-scale build and testing**

Bringing the cool environment to the worker.

Thank you • Questions?